Introduction

Prior studies show that smoking is linked to sharply increased long-term healthcare costs, mainly due to elevated risks of cardiovascular disease, cancer, and respiratory illness. BMI is also a strong determinant of chronic disease and, therefore, medical expenditure. Ward et al. (2021) found that higher BMIs are consistently associated with substantially greater per person healthcare spending in U.S. adults.

The *Medical Cost Personal Dataset* contains 1,338 observations and 7 variables. The dependent variable is the cost of a person's annual medical insurance. The 6 independent variables are age (ranging from 18 to 64), sex, body mass index (BMI), the number of children they have, whether they smoke, and the region they are from (Northeast, Northwest, Southeast, and Southwest). The dataset is composed of 676 males (50.5%) and 662 females (49.5%). 274 (20.5%) out of the 1,338 are smokers, meaning 1,064 (79.5%) are non-smokers. And, the data is fairly balanced across four U.S. regions.

This analysis has multiple objectives. First, to identify predictors that are significantly associated with the cost of insurance. Second, to model assumptions and determine if linear regression is appropriate and able to form an accurate representation of the dataset. Third, to see if nonlinear relationships exist between the independent and dependent variables. And finally, to compare the cost of insurance between smokers and non-smokers.

Two hypotheses were formed to investigate the dataset. The first hypothesis focuses on the mean medical insurance costs between smokers and non-smokers. The null hypothesis states the mean medical insurance for smokers and non-smokers are equal, while the alternative hypothesis states that the means are different. To test this hypothesis, a two-sample t-test was conducted to compare the average medical insurance costs between the two groups.

$$H_0: \mu_{smoker} = \mu_{non-smoker}$$
$$H_1: \mu_{smoker} \neq \mu_{non-smoker}$$

The second hypothesis explores if the independent variables have nonlinear effects on medical insurance costs. The null hypothesis states that nonlinear terms do not significantly improve model fit. The alternative hypothesis states that at least one term is nonlinear, and it significantly improves model fit. To test this hypothesis, a generalized additive model (GAM) with smooth terms for age and BMI was fitted and compared to the linear model to assess whether the inclusion of nonlinear effects improved the model representation of the dataset.

The linear model is expressed as
$log(charges) = B_0 + B_1 * age + B_2 * BMI + B_3 * smoker.$

The nonlinear model is expressed as
$log(charges) = B_0 + f_1(age) + f_2(BMI) + B_3 * smoker$, where f_1 and f_2 are nonlinear functions.

Methodology

The dataset was checked for missing values using the *.isna()* function, and no missing values were found. *Charges*, the dependent variable, has a right-skewed distribution, which was normalized by transforming it with *log()*. The dataset was also checked for outliers by using boxplots that plotted independent variables against *Log*(Charges). Extreme values were retained because they represent real-world high-cost cases. The independent variables sex, smoking status, and regional locale were converted into categorical factors to facilitate analysis. To address that right-skewed distribution of medical insurance charges, a log transformation was applied. Additionally, the variables BMI and smoking status were combined to explore joint patterns between the two.

The *summary()* function was used to assess fitted models. The *anova()* function was used for hypothesis testing. Model assumptions were evaluated using diagnostic plots. The functions *plot(, which =1)* and *plot(, which =3)* were used to check homoscedasticity and linearity. The function *plot(, which =2)* was used to check normality. And for independence, the function *dwtest()* was applied. The function *gam.check()* was used to verify assumptions for the GAM model. And last, the function *vif()* was used to detect potential multicollinearity among the independent variables.

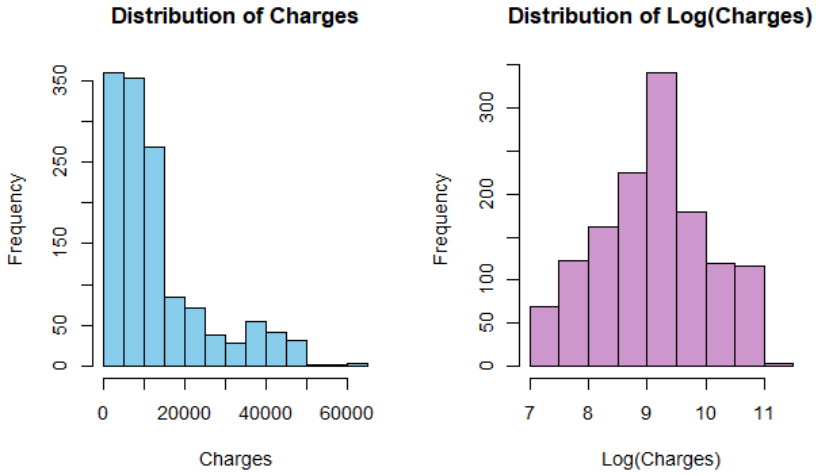
Exploratory Data Analysis

Figure 1: Summary of Variables

Variable	Minimum	1st Quarter	Median	Mean	3rd Quarter	Maximum
Age	18	27	39	39.21	51	64
BMI	15.96	26.30	30.40	30.66	34.69	53.13
Num. of Children	0.00	0.00	1.00	1.095	2.00	5.00
Charges	1122	4740	9382	13270	16640	63770

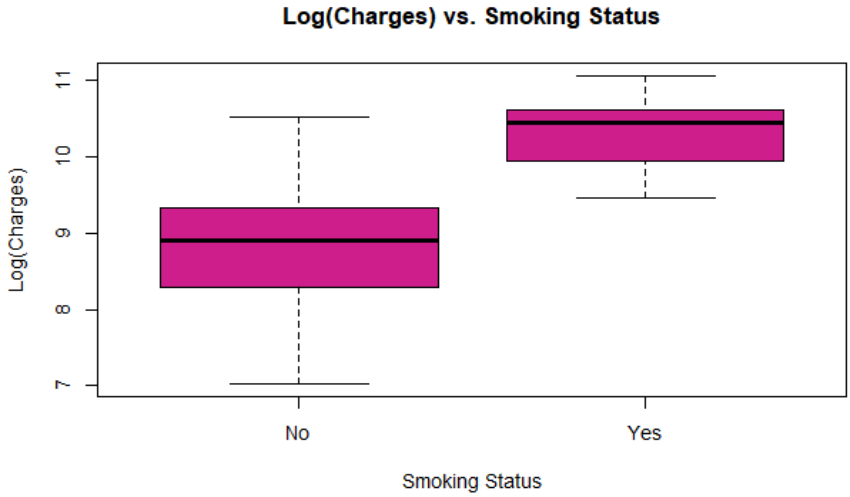
Log(Charges)	7.023	8.464	9.147	9.099	9.720	11.063

Figure 2: Histograms of Charges and *Log*(Charges)



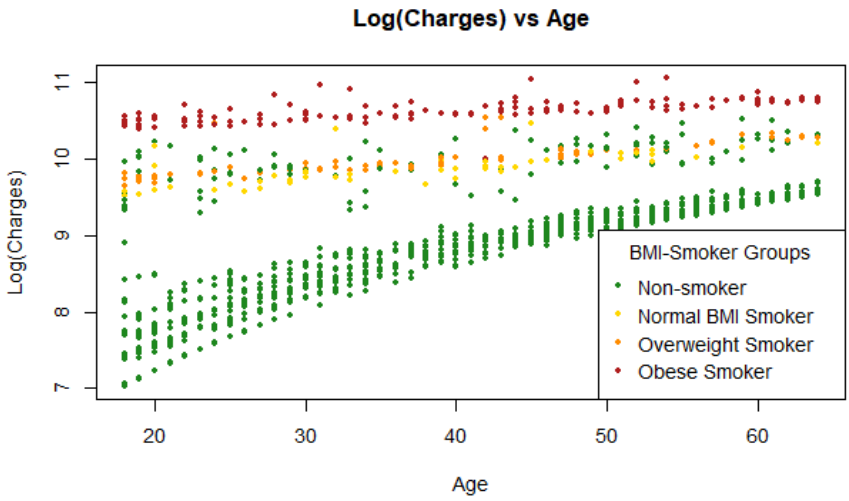
The original distribution of charges was right-skewed. By using *Log*(Charges), the distribution transforms into a more symmetrical/normal one, which backs this choice.

Figure 3: Boxplot of *Log*(Charges) by Smoking Status



Smokers have a higher median and interquartile range for *Log*(Charges) than non-smokers do. This indicates that smoking status has a strong effect on the cost of medical insurance.

Figure 4: Log(Charges) vs. Age Scatterplot Colored by BMI-Smoker Groups



There are three distinct layers in Figure 4. The color-coding reveals that the lowest layer is composed of non-smokers, the middle layer is made up of normal BMI/overweight smokers, and the highest layer has obese smokers. The separation suggests we test interaction terms and nonlinear models.

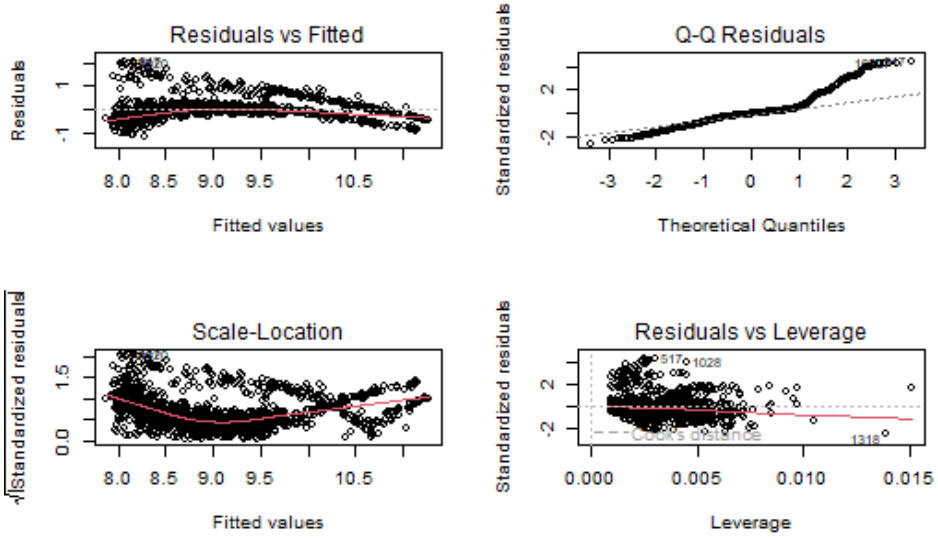
Results

Addictive Linear Regression Model:

$$logcharges = 7.0738 + 0.0351(age) + 0.0108(bmi) + 1.5459(smoker::Yes)$$

When *bmi* is held constant, with each year increase in *age*, there is roughly a 4% increase in *logcharges*. When *age* is held constant, with every unit increase in *bmi*, there is roughly a 1% increase in *logcharges*. When both *age and bmi* are held constant, the baseline charge for smokers is 371% higher than that of non-smokers. This means that two people of the same BMI and age can have wildly different medical insurance bills. All of the predictors are statistically significant, and the R^2 value is 0.744.

There was no violation of multicollinearity in this model. The VIF was approximately 1 for all variables. However, based on plotting and statistical tests, the model violates linearity, normality, and homoscedasticity assumptions. The Residuals vs. Fitted plot shows curvature and the QQ plot deviates from the diagonal. After removing the outlier (observation 517) and the leverage points (observations 1048 and 1318), assumptions did not improve.



Interaction Linear Regression Model:

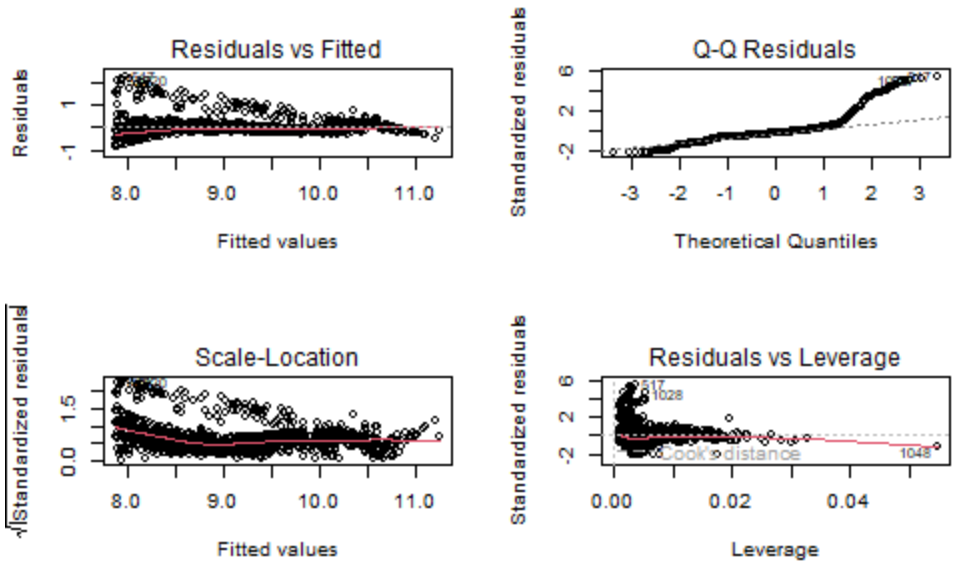
$logcharges = 7.1554 + 0.04205(age) - 0.00076(bmi) + 1.3185(smoker) - 0.04901(bmi * smoker) - 0.33(age * smoker)$

The interaction model can be broken down into two sub-models by assigning those that smoke with the value 1 and those that do not smoke with the value 0. Through dummy coding, the coefficients show how smoking changes the cost of insurance by modifying the effects of age and BMI. Here are the two separate models:

- Non-Smoker Model: $logcharges = 7.1554 + 0.04205(age) - 0.00076(bmi)$
- Smoker Model: $logcharges = 8.4739 + 0.00905(age) + 0.04825(bmi)$

Comparing these two sub-models, we see that smokers start at a higher intercept than non-smokers. Age has a greater impact on non-smokers' medical insurance costs compared to smokers' medical insurance costs. BMI has a greater impact on smokers' medical insurance costs than non-smokers' medical insurance costs. The R-squared value of this model is 0.7987, which is better than the initial linear model.

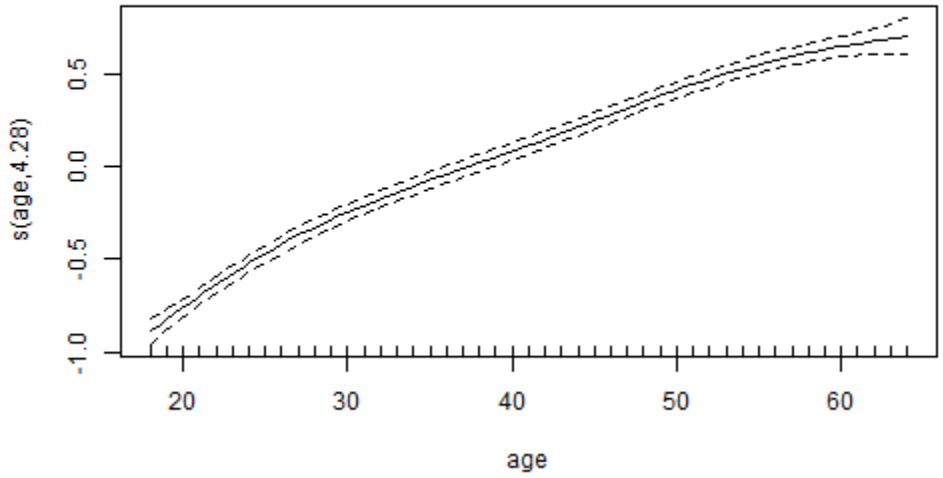
BMI is the only non-significant predictor. However, this doesn't matter much because the interaction term, *bmi*smoker*, is significant. Both the interaction terms are significant. Still, this model failed linearity, normality, and homoscedasticity assumptions, which might mean that non-linear models might represent the dataset more accurately.



Generalized Additive Model (GAM):

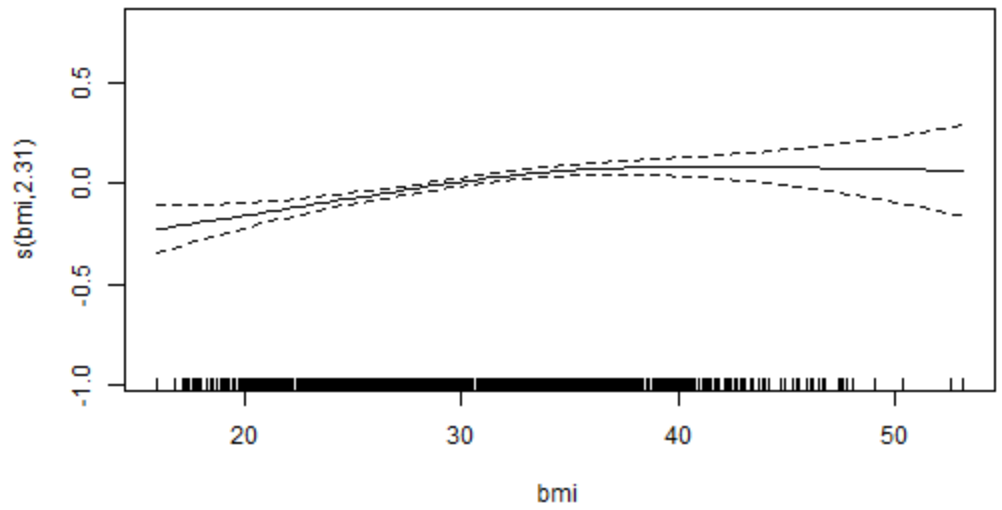
$$logcharges = 8.78178 + s(age) + s(bmi) + 1.54736 * smoker$$

Smoker, *s(age)*, and *s(bmi)* are statistically significant, indicating the effects of age and BMI on *log(charges)* are not linear. *s(age)* has an EDF of 4.28, which means moderate curvature, and *s(bmi)* has an EDF of 2.31, which indicates mild curvature. The model has an R-squared value of 0.752, which is higher than the additive linear model (0.744), but lower than the interaction model (0.7987). However, when tested for prediction, the GAM model had a lower RMSE compared to the interaction model, suggesting that the latter might be overfitting.



The plot above is for smoothed age. Younger individuals have a strong negative partial effect, meaning their predicted *log(charges)* are well below average. This partial effect increases rapidly between ages 20 and 40, then continues to gradually increase up to age 60. This means age

affects *log(charges)* nonlinearly, with a steeper rise in costs for younger people, followed by a shallower rise later into adulthood.



The plot above is of smoothed BMI. *log(charges)* increases as BMI rises from 20 to 35, but then levels off at higher BMI values. The leveling off could be attributed to the limited number of observations with extremely high BMI.

Using the gam.fit() function, this model was tested to see if it passed its own model diagnostics. Smoothed variables satisfied the base dimension test, which is a necessary test to run GAM models. This means that the model has the flexibility to capture nonlinear patterns. The QQ plot showed slightly better results than the linear models, but, the Residuals vs. Fitted plot showed heteroscedasticity. Overall, the GAM model satisfied the assumptions better than the previous two models, but the persistence of heteroscedasticity might mean there is a better model to use.

Hypothesis Testing

The first hypothesis examines whether smokers and non-smokers have the same average medical insurance charges. The null hypothesis states that there is no significant difference in mean insurance costs between the two groups. After conducting the statistical test, we found that the *p*-value is less than 2.2e-16, which is significantly lower than the α level. Therefore, we reject the null hypothesis and conclude that smokers and non-smokers have significantly different medical insurance costs. This results aligns with expectations as smoking is associated with increased health risks, which leads to increased medical costs.

The second hypothesis focuses on whether the smoothing terms add significant improvement to the model fit. The null hypothesis states that all smoothing terms are linear and do not contribute significantly to the model. To test whether nonlinear smoothing terms significantly improve the

model fit, an ANOVA table is used to compare a linear and a nonlinear model. The ANOVA table produced an F-statistic of 10.346 and a corresponding *p*-value of 3.868e-9, which allows us to reject the null hypothesis. This result indicates that at least one smoothing term is nonlinear and contributes significantly to improving the model's predictive accuracy. The relationship between the dependent variables and medical insurance charges cannot be adequately represented by a linear function. Nonlinear terms are needed to capture the complexity of the data.

Conclusion

This study finds that smoking, age and BMI are major determinants of how much medical insurance costs. Smokers consistently show substantially higher costs, and both age and BMI contribute positively to cost. However, all linear models fail key assumptions, including normality, homoscedasticity, and linearity. These failures persist even after a log transformation. The strong nonlinear patterns revealed by the GAM model and the presence of distinct smoker and non-smoker groups show that a single linear model cannot adequately capture these dynamics. For realistic interpretation, more flexible approaches would be more appropriate.

Health expenditures vary significantly between individuals, and understanding these patterns are important. Our results show that interaction terms involving smoking are statistically significant, indicating that smokers and non-smokers follow different cost structures. The incorporation of health-related variables could lead to more tailored insights.

References

Dourado, Raquel. Statistics with R. Kaggle, https://www.kaggle.com/code/goldens/statistics-with-r. Accessed 17 Nov. 2025.

Ward, Z. J., et al. “Association of Body Mass Index with Health Care Expenditures: Results from a Nationally Representative Sample of U.S. Adults.” PLOS One, vol. 16, no. 8, 2021, https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0247307. Accessed 17 Nov. 2025.